1 Introduction

Ledrappier's 1999 paper *Distribution des orbites des réseaux sur le plan réel* [ref] proves an ergodic theorem for averages of $\mathrm{SL}(2,\mathbb{R})$ orbits of real-valued functions. The goals of this document are twofold *(i)* rewrite Ledrappier's paper in the language of Teichmüller theory, to better motivate and understand the result, and *(ii)* to explicate terse elements of Ledrappier's proof.

The following notation will be used throughout

- $\mathbb{T}^2$ will denote the (topological) torus
- $\mathbf{T_1}$ will denote the Teichmüller space of $\mathbb{T}^2$
- Γ will denote the mapping class group of $\mathbb{T}^2$
- $\mathscr{Q}^1$ will denote the space of unit quadratic differentials over $\mathbf{T_1}$
- $\mathscr{MF}$ will denote the space of measured foliations on $\mathbb{T}^2$.
- Fix once and for all curves α, β with homology classes giving a basis for $\mathbb{T}^2$ and $i(\alpha,\beta) = 1$. Moreover, we may fix once and for all a covering map $\mathbb{R}^2 \to \mathbb{T}^2$, with (α,β) corresponding to the ordered lattice generators $((1,0),(0,1))$.

2 Teichmüller Theory for the Torus

Definition 1. *The Teichmüller space of the torus $\mathbf{T_1}$ will be defined as the set of pairs (S,φ) where S is a torus with a flat metric, and $\varphi : \mathbb{T}^2 \to S$ is an orientation preserving diffeomorphism, two pairs $(S_1,\varphi_1),(S_2,\varphi_2)$ will be identified if there is an isometry $I : S_1 \to S_2$, such that $I\varphi_1$ is homotopic to φ_2. By the uniformization theorem we may equivelantly consider Riemann surfaces and markings associated by homotopically trivial biholomorphism.*

In order to work with Teichmuller space, I will identify it with the space of abelian differentials of the form $dx + \tau dy$ for $\tau \in \mathbb{H}$, I will do this by first identifying $\mathbf{T_1}$ with representatives of the form $\mathbb{C}/(\mathbb{Z}+\tau\mathbb{Z})$, then pulling back the flat structure on the torus, by using the fixed covering on $\mathbb{T}$.

Start with a point $(S,\varphi) \in \mathbf{T_1}$, then we get an ordered pair of $\pi_1(S)$ generators from our marking $\varphi_*(\alpha)$ and $\varphi_*(\beta)$. Considering the covering map $\mathbb{C} \to S$, we can consider $A = 0 \cdot \varphi_*(\alpha), B = 0 \cdot \varphi_*(\beta)$ so that S is isometric to $\mathbb{C}/(A\mathbb{Z}+B\mathbb{Z})$. Such an identification with lattices is only defined up to linear isometry in $\mathbb{C}$, thus only up to rotation and dilation.

Conversely, we may start with a flat torus $S = \mathbb{C}/(A\mathbb{Z}+B\mathbb{Z})$, with $\pi_1(S)$ generated by α', β' with deck transformations giving A, B as above. There is a diffeomorphism $\varphi : \mathbb{T}^2 \to S$ with $\varphi_* : \alpha,\beta \mapsto \alpha',\beta'$, thus identifying $\mathbb{C}/(A\mathbb{Z}+B\mathbb{Z})$ with a point in Teichmüller space. Isometries of lattices homotopic to the identity must be linear transformations acting trivially on $H_1(S,\mathbb{Z})$, thus two marked lattices are identified with the same point in $\mathbf{T_1}$ exactly when they differ by a rotation and dilation.

We can choose a unique representative of each set of lattices up to rotation and dilation by multiplying by complex number A^{-1}, thus identifying our lattices with $\mathbb{C}/(\mathbb{Z}+\tau\mathbb{Z})$, for $\mathbb{Z} = A^{-1}B$. Since $\pi_1(S)$ acts by translation it is orientation preserving, thus the orientation on the universal cover $\mathbb{C}$ induces an orientation on S, thus our marking φ is orientation preserving when the pair of homology classes $\varphi_*[\alpha], \varphi_*[\beta]$ agrees with this induced orientation, this is exactly when $A^{-1}B \in \mathbb{H}$.[1]

Given a point in $(S,\varphi) \in \mathbf{T_1}$, now identified with $(\mathbb{C}/(\mathbb{Z}+\tau\mathbb{Z}), \varphi')$, there is a canonical abelian differential given by dz. Using our fixed covering $\mathbb{R}^2 \to \mathbb{T}^2$, we find that $\varphi'^*(dz) = dx + \tau dy$ our desired abelian differential. Moreover, this gives a natural identification $\mathbf{T_1} \longleftrightarrow \mathbb{H}$.

Definition 2. *A quadratic differential on a Riemann surface S is a section of the symmetric square of the holomorphic cotangent bundle $T^{1,0} \otimes T^{1,0}$. For a quadratic differential q, its norm is defined as $||q|| = \int_S |q|$, and the space of unit quadratic differentials on S is $\mathcal{Q}^1(S)$ is the set of quadratic differentials with norm 1. We define $\mathcal{Q}^1 = \{(S,\varphi,q) \mid (S,\varphi) \in \mathbf{T_1}\}$ and $q \in \mathcal{Q}^1(S)$.*

Consider a point $(S,\varphi) \in \mathbf{T_1}$, the coordinate charts on S are given by translation, thus for any two coordinates we have some constant so that $z_\alpha = z_\beta + c$. A holomorphic quadratic differential has a local expression $\phi_\alpha(z_\alpha)(dz_\alpha)^2$, and the transition laws for $T^{1,0} \otimes T^{1,0}$ are $\phi_\beta(z_\beta) = \phi_\alpha(z_\alpha)\left(\frac{dz_\alpha}{dz_\beta}\right)^2$, thus $\phi_\alpha(z_\alpha) = \phi_\beta(z_\beta)$ and since S is compact the maximum principle implies $\phi \in \mathbb{C}$ is a constant. Now for $\phi(dz)^2$ to be a unit quadratic differential, we require $|\phi| = \Im(\tau)^{-1}$ so that

$$1 = \int_S |q| = |\phi| \int_S |dz|^2 = |\phi| \, \mathrm{Area}(S) = |\phi| \, \Im(\tau)$$

Identifying (S,φ) with its torus representative as above, and using our fixed covering for $\mathbb{T}^2$, we can associate each quadratic differential q to $\varphi^*(q) = \frac{e^{i\theta}}{\Im(\tau)}(dx+\tau dy)^2$, note that this gives a natural identification of $\mathcal{Q}^1 \longleftrightarrow T^1\mathbb{H}$ since each quadratic differential is uniquely determined by (θ,φ). Since quadratic differentials on the torus are nonvanishing, we may take a locally well defined square root function, which allows us to associate 1-forms to q by considering $\Re(\sqrt{q}), \Im(\sqrt{q})$, where $\sqrt{q} = \sqrt{\phi}dz$. These naturally give rise to a foliation of the torus by considering their level sets, as well as a isotopy invariant measure for transverse curves by integrating their variation.

Definition 3. *A measured foliation of the torus is a foliation $\mathcal{F}$ of $\mathbb{T}^2$ into curves, equipped with a transverse measure μ on curves transverse to $\mathcal{F}$ satisfying μ is a locally finite borel measure on transverse curves and if I is an isotopy of curves such that $I([a,b],t)$ is transverse to $\mathcal{F}$ for all t, then $\mu(I([a,b],-))$ is*

[1] The identification of $\mathbf{T_1}$ with lattices follows roughly [ref FM]

constant. We identify two measured foliations if they are related by a homotopy of the torus.

Definition 4. *We can define the space of projectivized measured foliations* $\mathbb{P}\mathscr{M}\mathscr{F}$ *as* $\mathscr{M}\mathscr{F}/\mathbb{R}^\times$ *where* $\mathbb{R}^\times$ *acts via* $\lambda \cdot (\mathcal{F}, \mu) = (\mathcal{F}, \lambda\mu)$

Measured foliations on $\mathbb{T}^2$ may be associated with $(H^1(\mathbb{T}^2;\mathbb{R}) \setminus \{0\}/\pm) \cong \mathbb{RP}^1 \times \mathbb{R}_{>0}$. To see this, we take a basis η, ξ for $H^1(\mathbb{T}^2;\mathbb{R})$ satsfying

$$\int_\alpha \eta = 1 \quad \int_\beta \eta = 0 \quad \int_\alpha \xi = 0 \quad \int_\beta \xi = 1$$

Then we associate $\mathcal{F}$ to the unique (up to sign) cohomology class ω satisfying

$$i(\mathcal{F}, n\alpha + m\beta) = \left| \int_{n\alpha+m\beta} \omega \right|$$

writing ω in our basis, we associate $\mathcal{F}$ to $|b\eta + a\xi|$ for some $a, b \in \mathbb{R}$, similarly any nonzero cohomology class considered up to sign recovers a measured foliation $(\mathcal{F}, \mu)$ since by definition the leaves of our foliation are given by $\ker|\omega|$, with transverse measure $|\omega|$. Using our fixed covering $\mathbb{R}^2 \to \mathbb{T}^2$, we identify $(\eta, \xi) \longleftrightarrow (dx, dy)$, so that compatible with our notion of pullback for abelian and quadratic differentials each measured foliation can be written as $bdx + ady$ for some $a, b \in \mathbb{R}$. We also fix an isomorphism of $\mathscr{M}\mathscr{F} = (H^1(\mathbb{T}^2;\mathbb{R}) \setminus \{0\}/\pm)$ with $(\mathbb{RP}^1 \times \mathbb{R}_{>0})$, this is done by taking $(a, b) \mapsto b\eta + a\xi$, which we identify with $bdx + ady$, we will see later that this convention gives $\mathrm{SL}(2, \mathbb{R})$ equivariance and thus is the correct convention to agree with Ledrappier's paper [ref]. Under this identification for $X \in \mathscr{M}\mathscr{F}$ we find $[X] \in \mathbb{P}\mathscr{M}\mathscr{F}$ is the projection to the $\mathbb{RP}^1$ coordinate. [add more detail here about notation and boundary of Teichmuller space]

Definition 5. *The mapping class group of the torus is the set of orientation preserving diffeomorphisms* $\mathbb{T}^2 \to \mathbb{T}^2$ *with the relation* $\gamma \sim \gamma'$ *when they are homotopic. We denote the mapping class group of the torus as* Γ*.*

The mapping class group acts naturally on $\mathbf{T_1}, \mathscr{M}\mathscr{F}$, and $\mathcal{Q}^1$, these actions are respectively described as $\gamma : (S, \varphi) \mapsto (\gamma(S), \gamma\varphi)$ and the latter as $(\mathcal{F}, \mu) \mapsto (\gamma(\mathcal{F}), \gamma_*\mu)$, where for a transverse curve c, $\gamma_*\mu(c) = \mu(\gamma^{-1}(c))$ is the pushforward measure, and finally for a quadratic differential $q = \phi(dz)^2$ over a surface $S \in \mathbf{T_1}$ we simply pullback the differential $((S, \varphi), q) \mapsto (\gamma \cdot (S, \varphi), \phi(\gamma^* dz)^2)$. In order to work with measured foliations and relate them to unit quadratic differentials, we would like to use the cohomological description, thus we would like to compute explicitly the Γ action on $H^1(\mathbb{T}^2;\mathbb{R})$.

Since mapping classes on the torus are defined up to homotopy, a mapping class may be written as a map on $H_1(\mathbb{T}^2;\mathbb{Z})$, since these are invertible and orientation preserving, the resulting matrix lies in $\mathrm{SL}(2, \mathbb{Z})$. Mapping class groups are in general generated by Dehn twists [ref Martelli 6.5.4]. A Dehn twist about a curve c is defined as taking a tubular neighborhood D with a (orientation preserving) diffeomorphism to $r : D \to S^1 \times [-1, 1]$, and a smooth bump function $\chi : [-1, 1] \mapsto [0, 1]$ which is zero near -1 and one near 1, then defining $\gamma : \mathbb{T}^2 \to \mathbb{T}^2$ via $x \mapsto r^{-1}(e^{2\pi i \chi} r(x))$ on D and identity elsewhere. From which we may identify the Dehn twist about α as the matrix $\begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$ and the dehn twist about β as $\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$, since these two matrices generate $\mathrm{SL}(2, \mathbb{Z})$ we have thus identified Γ with $\mathrm{SL}(2, \mathbb{Z})$.

Now, given a matrix $g \in \mathrm{SL}(2, \mathbb{Z})$, which is associated to $\gamma \in \Gamma$, then γ acts on $H^1(\mathbb{T}^2; \mathbb{R})$ via g^{T}, which is seen from the following duality relation, which is well defined since integration is defined up to homology classes by Cauchy's theorem.

$$\int_c \gamma^* \omega = \int_{S^1} c^* \gamma^* \omega = \int_{S^1} (\gamma c)^* \omega = \int_{\gamma_* c} \omega$$

Since we have associated both $\mathbf{T_1}$ and $\mathscr{MF}$ to cohomology classes, this definition suffices to describe both mapping class group actions since the $\mathbf{T_1}$ action takes $\varphi^* dz \mapsto \varphi^* \gamma^* dz$, agreement for measured foliations is a consequence of the duality relation. The pullback of cohomology classes agrees with taking the pullback of the quadratic differential by definition.

3 Teichmuller Metric and Dilations

[add some exposition here]

Definition 6. *Given $q \in \mathcal{Q}^1$ lying above $(S, \varphi) \in \mathbf{T_1}$ we say that local coordinates (x, y) are natural coordinates when the horizontal and vertical foliations are given by $|dy|$ and $|dx|$ respectively.*

Existence of natural coordinates on the torus is obvious, since given the lattice presentation of a torus, and $q = \phi(dz)^2$ we simply need to multiply the lattice coordinates by $\phi^{-\frac{1}{2}}$ since our differential has no zeroes we can take such a well defined branch of the square root function. This allows us to define Teichmüller mappings, which are mappings that shear a surface along a pair of foliations.

Definition 7. *A map $h : S \to S'$ of Riemann surfaces is called Teichmüller when there are quadratic differentials q, q' on S, S' respectively such that for some $K \in \mathbb{R}_{>0}$ there are natural coordinates (x, y) near any point p and (x', y') near p' such that*

$$h(x + iy) = \sqrt{K} x' + \frac{i}{\sqrt{K}} y'$$

Such a map is said to have horizontal stretch factor K and dilation $\max\{K, K^{-1}\}$.

By Teichmüller's theorem [ref FM] every homotopy class of maps has a unique Teichmüller mapping, given two $(S, \varphi), (S', \varphi') \in \mathbf{T_1}$ the homotopy class is always taken to be the change of markings $\varphi' \varphi^{-1}$. This allows us to define a distance function on Teichmüller space, in terms of the dilation of a Teichmüller mapping between the marked surfaces.

Definition 8. *The Teichmüller distance between two surfaces is defined as*

$$d_{\text{Teich}}(S, S') = \frac{1}{2} \log K$$

Where K is the dilation of a Teichmüller map in the homotopy class of the change of markings.

The Teichmüller distance gives a complete metric on $\mathbf{T_1}$ and the geodesics are in bijection with $\mathcal{Q}^1$, this is defined as taking $q \in \mathcal{Q}^1$ with natural coordinates (x, y), then the Teichmuller geodesic $c(t)$ is given by taking the unit quadratic differential with natural coordinates $(t^{-1}x, ty)$. [ref FM].

Earlier we discussed how the $\mathbb{RP}^1$ coordinate of the measured foliation specifies a projective measured foliation, with data equivalent to the slope of a line on the torus. These projective measured foliations may be rigorously identified with $\partial \mathbf{T_1}$ in a way that is compatible with the notion of Teichmüller geodesics. For the purposes of understanding Ledrappier's paper, we can simply think of the topology on $\mathbf{T_1}$ as being defined by the Teichmüller metric. However, to understand this phenomena it is better to use a topology on the Teichmüller space given by embedding into the space of geodesic currents $\mathcal{C}$ with the weak-$*$ topology, as this is somewhat tangential I will not provide the details, and the reader is referred to [ref Martelli] for the proof in the negative Euler characteristic case. The embedding on the torus is given by taking (S, φ) to the functional on closed geodesics $c \mapsto \ell_S(\varphi(c))$ the length of the curve in S. We similarly have that measured foliations have a natural mapping to the space of currents, by taking $\mathcal{F}$ given by $|\omega|$ to $c \mapsto \int_c |\omega|$. Now we can further consider the projection $\mathcal{C} \to \mathbb{P}\mathcal{C}$ which will allow us to identify

Our goal is to understand the distribution of mapping class group orbits under the horocycle flow, which is related to the amount of shear a mapping class group element applies. Thus we will measure the size of a mapping class group element by the amount of shearing it applies to a surface.

Definition 9. *For $\gamma \in \Gamma$ we define K_γ to be the dilation of the Teichmüller map $h : S \to \gamma(S)$ for $(S, \varphi) \in \mathbf{T_1}$. This is well defined since Γ acts on $(\mathbf{T_1}, d_{\text{Teich}})$ by isometry which is immediate since the Teichmüller map is now between coordinate charts composed with the same biholomorphism.*

Finally we define a section from $\mathscr{MF}$ to $\mathcal{Q}^1$, using our fixed covering of $\mathbb{T}^2$ as a basepoint, then for $\mathcal{F} \in \mathscr{MF}$ we will find that there is a unique $\mathcal{F}'$ with $[\mathcal{F}'] = [\mathcal{F}] \in \mathbb{P}\mathscr{MF}$ so that $\mathcal{F}'$ is the horizontal foliation for some $q \in \mathcal{Q}^1$ lying above dz. Thus we can recover a point in $\mathcal{Q}^1$ by travelling along the Teichmüller line associated to q until we get a quadratic differential with horizontal foliation

$\mathcal{F}$. As discussed previously each quadratic differential is uniquely associated to one of the form $\frac{e^{i\theta}}{\Im\tau}(dx+\tau dy)^2$, thus fixing $\tau = i$ gives

$$\Im(\sqrt{q}) = \sin(\theta/2)dx + \cos(\theta/2)dy$$
$$\Re(\sqrt{q}) = \cos(\theta/2)dx - \sin(\theta/2)dy$$

Thus for the measured foliation associated to $X = (a,b)$, we consider $q \in \mathcal{Q}^1$ with $\mathcal{F}$ with horizontal foliation $||X||^{-1}(bdx + ady)$ and vertical foliation $||X||^{-1}(adx - bdy)$, then shearing gives us a quadratic differential q with horizontal and vertical foliations given respectively by $bdx+ady$ and $||X||^{-2}(adx-bdy)$. So that the resulting $q \in \mathcal{Q}^1$ satisfies

$$\sqrt{q} = (a||X||^{-2} + ib)\left(dx + \frac{ai - ||X||^{-2}b}{bi + ||X||^{-2}a}dy\right)$$

Definition 10. *Written explicitly we get*

$$\Psi : \mathbb{RP}^1 \times \mathbb{R}_{>0} \to \mathcal{Q}^1$$
$$(a,b) \mapsto (a||X||^{-2} + ib)^2 \left(dx + \frac{ai - ||X||^{-2}b}{bi + ||X||^{-2}a}dy\right)^2$$

The last preliminary notion to understand is the Teichmüller horocycle flow, however it will be far more illustrative to understand this as the the unipotent component of the $\mathrm{SL}(2,\mathbb{R})$ action on $\mathcal{Q}^1$. Let $g \in \mathrm{SL}(2,\mathbb{R})$, then the action of g on $\mathcal{Q}^1$ is simply the matrix action on the natural coordinates [ref– defined as in Forni's chapter 8 of dynamical systems handbook vol2]. Let (x', y') be natural coordinates for quadratic differential q, then in order to keep conventions consistent, we take the action on natural coordinates as $g\begin{pmatrix} y' \\ x' \end{pmatrix}$, thus the action on cohomology classes is given by the pullback, so that the action is given explicitly as

$$\begin{pmatrix} \Im\sqrt{q\cdot g} \\ \Re\sqrt{q\cdot g} \end{pmatrix} = g^{\mathrm{T}} \begin{pmatrix} \Im\sqrt{q} \\ \Re\sqrt{q} \end{pmatrix}$$

The Teichmüller unipotent flow is given by the action of the unipotent subgroup $\left\{ h_s = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix} \mid s \in \mathbb{R} \right\}$, moreover this realizes the shearing used to define the map Ψ as the action of the diagonal subgroup. Also note that this gives a second identification of Γ with $\mathrm{SL}(2,\mathbb{Z})$.

The definition of the $\mathrm{SL}(2,\mathbb{R})$ action on $\mathcal{Q}^1$ induces one on $\mathbf{T_1}$ by projection, using our identification of $\mathbf{T_1}$ with $\mathbb{H}$ it is immediate that if q lies over the surface $dx + \tau dy$ that $q \cdot g$ lies over $dx + g\cdot\tau dy$ where g is taken to act on τ by mobius transform. Thus the $\mathrm{SL}(2,\mathbb{R})$ action on $\mathcal{Q}^1$ carries over to one on $T^1\mathbb{H}$. The final piece in being able to use the geometry of $\mathbb{H}$ to describe the Teichmüller dynamics is that $(\mathbf{T_1}, 2d_{\mathrm{Teich}})$ is isometric to $(\mathbb{H}, d_{\mathbb{H}})$, moreover this isometry respects the $\mathrm{SL}(2,\mathbb{R})$ action, in particular the Teichmüller unipotent flow is taken to the horocycle flow [ref FM].

Now we give a complete coordinate description of $\mathcal{Q}^1$ of the form $\mathscr{MF} \times \mathbb{R}$, decomposing into shearing about a measured foliation and the unipotent flow. This is defined as taking $(X, s) \mapsto \Psi(X) \cdot h_s$, to see this map gives a bijection write $X = g\begin{pmatrix}1\\0\end{pmatrix}$, for $g = R(\theta)D(||X||, ||X||^{-1})$ the composition of a diagonal and rotation matrix in $\mathrm{SL}(2, \mathbb{R})$. It follows that

$$\Psi(X) \cdot h_s = \Psi\left(g\begin{pmatrix}1\\0\end{pmatrix}\right) \cdot h_s = (dx + idy)^2 \cdot gh_s$$

Now under the identification with $T^1\mathbb{H}$ this gives gh_s acting on $(i, (0, 1))$, every $\mathrm{SL}(2, \mathbb{R})$ matrix has a unique decomposition of the form $R(\theta)D(\lambda, \lambda^{-1})h_s$, and since this action is free and faithful we get a bijection. We may write the map and its inverse explicitly in coordinates, by applying the Teichmüller unipotent flow to the horizontal and vertical foliations of $\Psi(X)$, which gives horizontal and vertical foliations $bdx + ady$ and $||X||^{-2}(adx - bdy) + s(bdx + ady)$. This corresponds to the quadratic differential q with

$$\sqrt{q} = (bs + a||X||^{-2} + ib)\left(dx + \frac{as - b||X||^{-2} + ia}{bs + a||X||^{-2} + ib}dy\right)$$

Using this closed form, given a quadratic differential of the form $q = \frac{e^{i\theta}}{\Im(\tau)}(dx + \tau dy)^2$, the preimage under our coordinate map is

$$\begin{aligned}
\Im(\tau) &= \frac{1}{b^2 + (a||X||^{-2} + bs)} \\
\Re(\tau) &= \Im(\tau)(abs^2 - ab||X||^{-4} + ab - b^2 s||X||^{-2} + a^2 s||X||^{-2}) \\
\theta &= 2\arg(bs + a||X||^2 + ib)
\end{aligned}$$

Definition 11. *Let $\gamma \in \Gamma$ and $X \in \mathbb{RP}^1 \times \mathbb{R}_{>0}$. Then we can define $s(\gamma, X)$ by the equation*

$$\Psi(\gamma X) = \gamma\Psi(X) \cdot h_{s(\gamma,X)}$$

[Need more justification here] [Show the inverse for $\Psi(X)h_s$ and thus that parameterizing by Ψ and s describes all of $\mathcal{Q}^1$]

4 Statement of Theorem

Let $f \in C_c(\mathbb{RP}^1 \times \mathbb{R}_{>0})$, and suppose that $[X] \in \mathbb{RP}^1$ has irrational slope, then

5 Proof of Theorem

Let $X \in \mathbb{RP}^1 \times \mathbb{R}_{>0}$ be a vector with irrational slope and let $f \in C_c(\mathbb{RP}^1 \times \mathbb{R}_{>0})$, then we can define a lift $\tilde{f}$ of f to $\mathcal{Q}^1$, by distributing f along the Teichmüller unipotent flow.

Definition 12. *Let $\epsilon > 0$, then let χ_ϵ be a continuous function on $\mathbb{R}$ supported on $(-\epsilon, \epsilon)$ with $\int_{\mathbb{R}} \chi_\epsilon = 1$. Then for $q \in \mathcal{Q}^1$, bijectivity of the coordinate construction gives a unique s, such that $q \cdot h_s$ lies in $\mathrm{Im}(\Psi)$ use this to define*

$$\tilde{f}(q) = f(\Psi^{-1}(q \cdot h_s))\chi_\epsilon(s)$$

We may further define π to be the quotient map $\mathcal{Q}^1 \to \mathcal{Q}^1/\Gamma$ and

$$\begin{aligned} \overline{f} &: \mathcal{Q}^1/\Gamma \to \mathbb{R} \\ S \cdot \Gamma &\mapsto \sum_{\Gamma} \tilde{f}(\pi(S)) \end{aligned}$$

By construction, integrating $\bar{f}$ along the unipotent flow gives the sum over $\{\gamma \in \Gamma \mid \gamma X \in \mathrm{supp}(f)\}$ with $s(\gamma, X)$ lying within the integral bounds up to some small error, by relating this to the average of $f(\gamma X)$ with K_γ small, we get a new formula for the sum from the equidistribution of the unipotent flow. We can use the hyperbolic geometry to relate these two quantities as follows

Lemma 1.

$$s(\gamma, X)^2 = \frac{K_\gamma + K_\gamma^{-1}}{||X||^2||\gamma X||^2} - \frac{1}{||X||^4} - \frac{1}{||\gamma X||^4}$$

Proof. [Copy paste proof here] □

Now since f has compact support, the $\frac{1}{||\gamma X||^4}$ term is bounded for $\gamma X \in \mathrm{supp}(f)$ and also note that $K_\gamma \geq 1$, so that the $\frac{K_\gamma^{-1}}{||X||^2||\gamma X||^2}$ is also uniformly bounded for all γ, this furnishes positive constants C and D such that

$$\frac{K_\gamma}{||X||^2||\gamma X||^2} - C \leq s(\gamma, X)^2 \leq \frac{K_\gamma}{||X||^2||\gamma X||^2} + D$$

whence we find that

$$\begin{aligned} |s(\gamma, X)| &\leq \frac{T}{||X||||\gamma X||} - \sqrt{C} \\ \implies \sqrt{K_\gamma} &\leq ||X||||\gamma X|| \left(\sqrt{\frac{K_\gamma}{||X||^2||\gamma X||^2} - C} + \sqrt{C} \right) \leq T \end{aligned}$$

and if $\sqrt{K_\gamma} \leq T$, then

$$|s(\gamma, X)| \leq \sqrt{\frac{T^2}{||X||^2||\gamma X||^2} + D} \leq \frac{T}{||X||||\gamma X||} + \sqrt{D}$$

The first thing to notice from these equations is that $\overline{f}$ is well defined, to see this we use that for $g \in \mathrm{SL}(2, \mathbb{Z})$ the matrix representation of γ

$$K_\gamma + \frac{1}{K_\gamma} = 2\cosh 2d_{\mathrm{Teich}}(S, S \cdot \gamma) = 2\cosh d_{\mathbb{H}}(i, \gamma \cdot i) = ||g||_F^2$$

So that for any N there are finitely many γ with $|s(\gamma, X)| \leq N$, this further implies $\bar{f}$ is continuous since it is given locally by a sum of continuous functions. [A more geometric way to see than relating to the matrix norm is that the mapping class group acts properly discontinuously]. Secondly defining $L = \inf_{Y \in \mathrm{supp}(f)} ||Y||$ and $M = \sup_{Y \in \mathrm{supp}(f)} ||Y||$ we get the following explicit relation for positive valued f,

$$\int_{-\frac{T}{||X||M}+\sqrt{D}+\epsilon}^{\frac{T}{||X||M}-\sqrt{D}-\epsilon} \overline{f}(\pi\Psi(X)\cdot h_s)ds \leq \sum_{\substack{\gamma\in\Gamma \\ \sqrt{K_\gamma}\leq T}} f(\gamma X) \leq \int_{-\frac{T}{||X||L}-\sqrt{C}-\epsilon}^{\frac{T}{||X||L}+\sqrt{C}+\epsilon} \overline{f}(\pi\Psi(X)\cdot h_s)ds$$

$$\lim_{T\to\infty}\frac{1}{T}\int_{-\frac{T}{||X||M}}^{\frac{T}{||X||M}} \overline{f}(\pi\Psi(X)\cdot h_s)ds \leq \lim_{T\to\infty}\frac{1}{T}\sum_{\substack{\gamma\in\Gamma \\ \sqrt{K_\gamma}\leq T}} f(\gamma X) \leq \lim_{T\to\infty}\frac{1}{T}\int_{-\frac{T}{||X||L}}^{\frac{T}{||X||L}} \overline{f}(\pi\Psi(X)\cdot h_s)ds$$

Similar equalities holds for negative valued f. We may assume that f is strictly positive or negative, considering f^+, f^- defined as the restrictions $f|_{\{f>0\}}, f|_{\{f<0\}}$ respectively. To complete the proof I will evaluate $\lim_{T\to\infty}\frac{1}{T}\int_{-\frac{T}{||X||M}}^{\frac{T}{||X||M}} \overline{f}(\pi\Psi(X)\cdot h_s)ds$, since the same argument applies to the other limit squeezing the sum and gives the same result.

[Need to check for aperiodicity] By Dani's theorem we have for μ the unit Haar measure on $\mathrm{SL}(2,\mathbb{R})/\mathrm{SL}(2,\mathbb{Z})$

$$\begin{aligned}\lim_{T\to\infty}\frac{1}{T}\int_{-\frac{T}{||X||M}}^{\frac{T}{||X||M}} \overline{f}(\pi\Psi(X)\cdot h_s)ds &= \lim_{T\to\infty}\frac{1}{T||X||M}\int_{-T}^{T} \overline{f}(\pi\Psi(X)\cdot h_s)ds \\ &= \frac{2}{||X||M}\int \overline{f}(\pi\Psi(X)\cdot g)d\mu\end{aligned}$$

Lemma 2.

$$\int \overline{f}(\pi\Psi(X)\cdot g)d\mu = \frac{6}{\pi^2}\int \tilde{f}\left(\Psi\begin{pmatrix} b \\ a\end{pmatrix}\right)\cdot h_s\,|db\,da\,ds|$$

Proof. We can parameterize $\mathrm{SL}(2,\mathbb{R})$ by $\left\{\begin{pmatrix} e & f \\ g & h\end{pmatrix} \mid eh - fg = 1\right\}$, then the integral over $\mathrm{SL}(2,\mathbb{R})$ is the integral over the chart $e \neq 0$. In this chart we have $d\mu = \frac{6}{\pi^2}\frac{|de\,df\,dg|}{|e|}$, in our earlier parameterization in terms of a, b, s we have $a = e, b = g, f = as - b(a^2+b^2)^{-1}$, this gives us Jacobian determinant a. Now letting D be a fundamental domain for $\mathrm{SL}(2,\mathbb{R})/\Gamma$ we get from definition of $\tilde{f}$

$$\begin{aligned}
\int_{\mathrm{SL}(2,\mathbb{R})/\Gamma} \overline{f}(\pi\Psi(X)\cdot g)d\mu &= \int_D \sum_{g'\in\Gamma} \tilde{f}(\Psi(X)\cdot g'g)d\mu \\
&= \int \tilde{f}\left(\Psi(X)\cdot\begin{pmatrix} e & f \\ g & h\end{pmatrix}\right)\frac{6}{\pi^2}\frac{de\,df\,dg}{|e|} \\
&= \frac{6}{\pi^2}\int \tilde{f}\left(\Psi(X)\cdot\begin{pmatrix} a & as-b||X||^{-2} \\ b & bs+a||X||^{-2}\end{pmatrix}\right)da\,db\,ds \\
&= \frac{6}{\pi^2}\int \tilde{f}\left(\Psi\begin{pmatrix} a \\ b\end{pmatrix}\cdot h_s\right)da\,db\,ds \\
&= \frac{6}{\pi^2}\int f(a,b)\chi_\epsilon(-s)da\,db\,ds \\
&= \frac{6}{\pi^2}\int f(a,b)da\,db \\
&= \frac{6}{\pi^2}\int_{\mathbb{RP}^1\times\mathbb{R}_{>0}} f(re^{i\theta})rdrd\theta
\end{aligned}$$

[fix integration domain a bit more] □

Combining results gives us

$$\lim_{T\to\infty}\frac{1}{T}\int_{-\frac{T}{||X||M}}^{\frac{T}{||X||M}} \overline{f}(\pi\Psi(X)\cdot h_s)ds = \frac{12}{\pi^2||X||M}\int_{\mathbb{RP}^1\times\mathbb{R}_{>0}} f(re^{i\theta})rdrd\theta$$

eliminate the $1/M$ factor, we note that f is supported on the annulus $\{re^{i\theta} \mid L\le r\le M\}$, we can take for any natural N, a continuous partition of unity of the annulus $\chi^1,\dots,\chi^N$ with χ^j supported on $\{re^{i\theta} \mid L_j = L+\frac{M-L}{N}(j-2) < r < L+\frac{M-L}{N}(j+1) = M_j\}$, writing $f^j = \chi^j f$ we recover that

$$\sum_1^N \lim_{T\to\infty}\frac{1}{T}\int_{-\frac{T}{||X||M_j}}^{\frac{T}{||X||M_j}} \overline{f^j}(\pi\Psi(X)\cdot h_s)ds \le \lim_{T\to\infty}\frac{1}{T}\sum_{\substack{\gamma\in\Gamma \\ \sqrt{K_\gamma}\le T}}\sum_j f^j(\gamma X) = \lim_{T\to\infty}\frac{1}{T}\sum_{\substack{\gamma\in\Gamma \\ \sqrt{K_\gamma}\le T}} f(\gamma X)$$

Applying Dani's theorem and lemma 2 to the term on the left hand side we get

$$\sum_1^N \lim_{T\to\infty}\frac{1}{T}\int_{-\frac{T}{||X||M_j}}^{\frac{T}{||X||M_j}} \overline{f^j}(\pi\Psi(X)\cdot h_s)ds = \sum_1^N \frac{12}{\pi^2||X||M_j}\int_{\mathbb{RP}^1\times\mathbb{R}_{>0}} f^j(re^{i\theta})rdrd\theta$$

Then since this hold for all N, we may write the following and take $N \to \infty$

$$\left|\frac{12}{\pi^2||X||}\int_{\mathbb{R}P^1\times\mathbb{R}_{>0}} fdrd\theta - \sum_1^N \frac{12}{\pi^2||X||M_j}\int_{\mathbb{RP}^1\times\mathbb{R}_{>0}} f^j rdrd\theta\right|$$
$$=\frac{12}{\pi^2||X||}\left|\int_{\mathbb{R}P^1\times\mathbb{R}_{>0}} \sum_1^N f^j - \frac{1}{M_j} f^j rdrd\theta\right|$$
$$\leq\frac{12||f||_\infty}{\pi^2||X||}\sum_1^N\int_0^\pi\int_{L_j}^{M_j} 1-\frac{L_j}{M_j}drd\theta \leq \frac{12||f||_\infty}{\pi||X||L}\sum_1^N (M_j-L_j)^2$$
$$\leq\frac{12||f||_\infty}{\pi||X||L}\frac{(M-L)^2}{N}$$

The exact same proof applies to the upper bound, so that we get the desired result

$$\frac{12}{\pi^2||X||}\int_{\mathbb{R}P^1\times\mathbb{R}_{>0}} fdrd\theta = \lim_{T\to\infty}\frac{1}{T}\sum_{\substack{\gamma\in\Gamma \\ \sqrt{K_\gamma}\leq T}} f(\gamma X)$$

6 References